

AUDIT_LEX_NOVI_AGENT_ASSETS.md

Audit & Extraction Report: Subfolder 1 of ___Lex-Novi-Æxentis-Copiæ (Agent-Assets&Global-Documentation_Vault)

Scope: Detailed domain audit, W5+H mapping, and architectural extractions from the external vendor vault.

Parent Source:

___Lex-Novi-Æxentis-Copiæ/Agent-Assets&Global-Documentation_Vault
(1ksP096a6y3IE11awDuUE79nNWsSYqpX4).

Status: All original files preserved 100% untouched.

1. Executive Summary & Core Purpose

Agent-Assets&Global-Documentation_Vault is the **Sensory Ground Truth (Tier 1)** and **Immutable External Reference Library** for all silicon, OS, and framework technologies powering the ecosystem.

It contains 13 dedicated domain subfolders and 4 root specifications describing external systems (Qualcomm, Google, Unsloth, Llama.cpp, and research foundations).

The Architectural Transition:

- **Historical Rule Cataloged:** The folder's **CLAUDE.md** contained the older four-artifact convention (**name.pdf** + **name.md** + **name.jsonl** + **skills.md**).
 - **Active Law Re-established:** This document confirms that the four-artifact convention was superseded by the **5+1 Cognitive Memory Architecture**, the **5 Manifest Document Types** (Tool, Skill, Reference, Memory, Data), and the **Living LLM Wiki** standard.
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2. Domain-by-Domain Audit & Inventory

Domain Subfolder	Item Count	Core Technical Assets Mined	Architectural Destination
#QAIRT_main	18 items	Qualcomm AI Engine Direct SDK, QairtApi, HTP SDK, GenieX runtime flags, LiteRT C++ bindings, ExecuTorch CLIP.	Enveloped into vendor-corpora/qualcomm-qairt-sdk/ & Spec 01.
GOOGLE-DEV-DOCS	10 items	NDK ASharedMemory , Abstract UNIX domain sockets (\0), LMK shielding (FOREGROUND_APP_ADJ), GCP cross-account handshakes, Gemma 4 deployment.	Enveloped into vendor-corpora/google-android-platform/ & Spec 01.
unsloth-docs	4 items	Unsloth local UI, Llama-server OpenAI endpoint deployment guides (PDF & MD), GGUF export parameters.	Enveloped into vendor-corpora/unsloth-docs/ & Spec 03.
llama-docs	2 items	llama.cpp-npu and native C/C++ inference engine documentation.	Enveloped into vendor-corpora/llama-docs/ & Spec 01.
FraQAT	12 items	Fractional Quantization-Aware Training research papers, full Markdown transcriptions, and JSONL data banks.	Enveloped into 03_recall_cache/ quantization specs.

Domain Subfolder	Item Count	Core Technical Assets Mined	Architectural Destination
Continual harness	1 item	87KB full-text canonical research paper: <i>Online Adaptation for Self-Improving Foundation Agents</i> .	Enveloped into Prime Agent Mode B specifications in Spec 03.
Snapdragon NPU LLM	6 items	Proven benchmarks running LLMs on Hexagon DSP (verified 31 tok/s on older v69; informs 90 tok/s on v79).	Ingested into 02_wiki_md/architectures/snapdragon_npu.md .
KV-CACHE_quantization	1 item	TurboQuant guide for llama.cpp, cutting attention KV-cache RAM footprint by up to 60%.	Ingested into 03_recall_cache/kv_store/ .
Readme-Notes	4 items	Historical readmes in .docx and .pdf formats.	Retained untouched in place for historical provenance.
Edge AI hub integration	Subfolder	Model compilation workflows for Qualcomm AI Hub.	Ingested into vendor-corpora/qualcomm-qairt-sdk/ .
@ skill manifest jsonl	Subfolder	Manifest conventions linking document artifacts on shared_basename .	Formalized into Master Spec 00 & Spec 02.
#GOOGLE_AGENTIC-AI	Subfolder	Vertex AI Enterprise Agent Platform guides (\$1,000 credit pipeline).	Ingested into cloud fallback routes in Spec 03.

Domain Subfolder	Item Count	Core Technical Assets Mined	Architectural Destination
Automated Build Android...	Subfolder	GitHub Actions configurations for building headless Android daemons.	Ingested into repository CI/CD templates.

3. Key Technical Insights Preserved

- Direct GGUF Inference on Qualcomm NPU (GenieX):**
 - Confirmed from #QAIRT_main: GenieX supports running GGUF Q4_0 directly on Snapdragon NPU via the GGML Hexagon backend without requiring offline x86 Linux compilation.
- Zero-Copy Shared Memory (Google NDK):**
 - Confirmed from GOOGLE-DEV-DOCS: ASharedMemory_create + mmap + abstract UNIX sockets (sun_path[0] = '\0') with SCM_RIGHTS bypasses Android's 1MB Binder IPC limit completely.
- Reset-Free Agent Self-Correction:**
 - Confirmed from Continual harness: Agents improve by updating prompt layers (/refine) over an immutable base prompt with automated rollback snapshots, eliminating catastrophic forgetting.